

Field demonstration of an unshielded InP photonic integrated circuit (PIC) onboard a PocketQube picosatellite payload

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Abstract We developed a picosatellite payload with a PIC consisting of a laser and a photodiode, mounted with no metallic shielding. Experimental field results demonstrate a success in the mission after more than one year orbiting the earth, showing no degradation due to ionizing radiation. ©2026 The Author(s)

Introduction

Recently, there has been an exponential increase in the number of small-satellite constellations, owing to a decrease in launching costs which enables the feasibility of affordable spatial missions, and the miniaturization of electronic components [Kulu2021]. Most of these low-cost satellites use a Low Earth Orbit (LEO), with altitudes ranging from 160 km to 2000 km. Applications of these small satellites include optical Earth observation, weather prediction, broadband internet and Internet of Things (IoT), among others [1–3].

Microwave photonics (MWP) is a promising technique for the generation of carrier waves in the millimeter-wave (30–300 GHz) and terahertz (300 GHz–3 THz) ranges, which is a key technology to enable large communication bandwidths [4]. Potential applications of MWP in space include transceivers for intra-satellite communications, signal up- and down-conversion, RF carrier generation, and others [5]. Furthermore, the miniaturization of photonic components into optical chips or photonic integrated circuits (PIC) is another key technology if mmW/THz radio communications are used in small satellites with form factors following the CubeSat or PocketQube standards [3]. Recently, there has been significant research interest in the use of PICs onboard small satellites, and the effects that the harsh space environmental conditions may produce in the reliability and long-term operation of such PICs [6,7].

In this work, we report on the design of a picosatellite payload which has a PIC designed for monolithic generation of data modulated millimeter-wave carrier waves [8]. This PIC consists of two lasers and a photodiode, of which only one laser is used in this work. The PIC was glued and wire-bonded to the PCB tracks with no encapsulation or shield protection against ionizing radiation. The picosatellite was launched on 15 March

2025, and the experimentally received telemetry data, collected over a period of more than one year, shows no degradation of the PIC operation with an average received optical power of -12.02 dBm over a temperature range of [5, 30] °C.

Satellite and mission description

Figure 1 shows a render image of the picosatellite. The picosatellite adheres to the 1.5P PocketQube standard, with a square-prismatic shape with an internal volume below 250 cm³. Most of this volume is occupied by the satellite essential subsystems, including the battery, the On-Board Computer (OBC), the communication module and the power management module. The picosatellite is powered by four 750 mW solar panels that charge a 2400 mAh lithium-ion battery. This battery provides a maximum output current of 1 A with an output voltage ranging from 3.2 V to 4.2 V. The designated payload space is located on one of the square bases of the prism, offering a usable volume of approximately 10.6 cm³ and supporting a maximum mass of 25 g with a maximum height of 7.7 mm [9]. A picture of the payload form factor can be shown on Fig. 1.

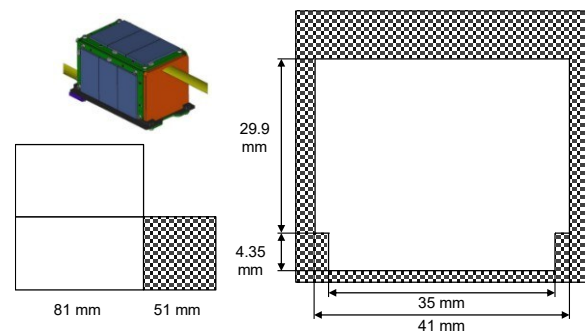


Fig. 1: A render of the picosatellite, dimensions of the satellite framework and of the available space for the developed payload. The payload PCB occupies one of the picosat side facets and is shown in red in the figure.

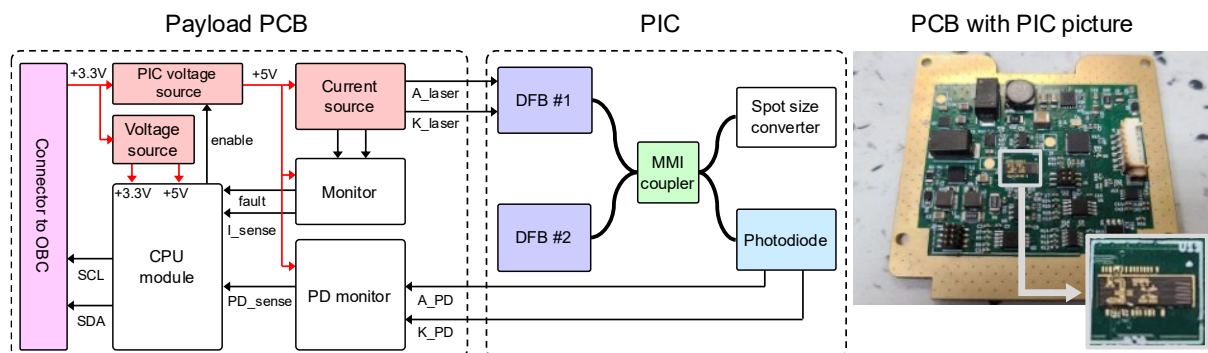


Fig. 2: Diagram of the payload circuitry, showing the main subsystems that compose the payload circuitry, and a diagram of the PIC and its integrated optical components. The picture on the left shows the flight-assembled PCB, with the PIC glued and wire-bonded to the PCB and located in the center of the assembly.

The OBC governs the operation of the satellite subsystems including the payload. The payload is unpowered most of the time to save battery power, and it is powered on if the following three conditions are met:

- The main battery off the satellite has enough charge.
- The current temperature of the satellite is in the range from 5 to 30 °C.
- More than 90 minutes—approximate orbital period—have passed since the last payload activation.

If the above conditions are met, the OBC powers on the payload and waits 5 seconds for stabilization. The OBC then sends a measurement command and polls the payload until the measurement is ready. The telemetry block consists of 32 bytes of data. This payload has a CRC32 which is used by the OBC and the ground station to check for errors. The telemetry block is stored in the OBC memory and transmitted to a ground station located in Madrid, Spain.

The picosatellite carrying the payload was deployed by a Falcon 9 vehicle on SpaceX Transporter-13 mission [10]. This mission was launched on 15 March 2025 at 06:43 UTC, from Space Launch Complex 4E (SLC-4E) in Vandenberg Space Force Base, California, USA. The nanosatellite was inserted in a LEO orbit, specifically a polar Sun-Synchronous Orbit (SSO), with an approximate altitude of 500 km and an orbital inclination of 97 degrees, with a mission life expectancy of more than 12 months. The satellite orbit, position and Earth line-of-sight visibility can be tracked in real-time on the SatNOGS online database with NORAD id 63490 [11].

Payload design, implementation and integration process

Figure 2 shows a block diagram of the developed payload. The payload is connected to the picosatellite power system which provides a regulated voltage of 3.3 V when the payload is

activated. The payload communicates with the OBC using the I2C protocol. In the payload, a fixed voltage regulator and converter provide 3.3 V and 5 V outputs to the CPU module, which is always activated when the payload is powered on. The payload CPU is an ATmega 328P microcontroller. A secondary voltage source, which is enabled by the payload CPU when a measurement is taking place, feeds a fixed current source of 60 mA to power the semiconductor optical amplifier (SOA) one of the two distributed feedback (DFB) lasers integrated into the PIC. A current-to-voltage converter is used to verify the correct operation of the laser, and if a laser malfunction is detected the OBC is notified to avoid wasting energy. The subsystem named PD monitor consists of a signal conditioning circuit to reverse-bias the integrated photodiode and convert the photocurrent to a voltage value that can be digitized by one of the microcontroller analog to digital converters (ADCs).

The PIC used in this mission is a monolithically integrated III-IV PIC composed of two tunable distributed feedback (DFB) lasers, a 2x2 multi-mode interference (MMI) coupler, a spot size converted and a PIN photodiode, as shown in Fig. 2. For more information about this PIC please refer to [8]. In this field demonstration, only one of the two available lasers was used.

A picture of the flight model of the developed payload PCB can be seen on the left side of Fig. 2. The PIC was glued to the PCB with epoxy resin and wire-bonded, with no metallic shielding or cover whatsoever, and is in the center of the assembly. The PCB features a white Molex connector on the left side for power and I2C communication with the satellite OBC.

Experimental results

The first telemetry block received by the ground station in Madrid has a timestamp of 31/03/2025, and until 08/04/2026, the time of writing of this manuscript, the total number of telemetry blocks

received is 1667. Figure 3 shows a histogram composed of the number of telemetry blocks received for each temperature value in the range from 5 to 30 °C. The nonuniformity could be explained by the orbit characteristics and the ground station location on Earth, and the fact that the satellite does not feature temperature control.

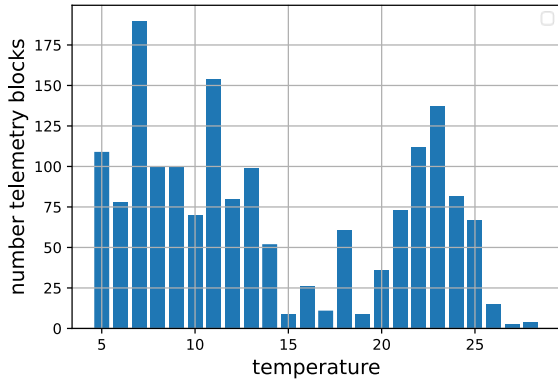


Fig. 3: Number of telemetry data blocks received for each temperature figure.

Figure 4 shows the time evolution of the measured photocurrent for a selected set of temperatures of 6, 10, 21 and 25 °C. No discernible variation can be observed for each temperature value, which suggests that there is no variation in the photocurrent over time and hence on the received optical power and laser operation. Moreover, we can assume that the photocurrent for each temperature value is a mean-ergodic, wide-sense stationary process, and hence we can take an estimation of the temporal average as the average of the process for each temperature value.

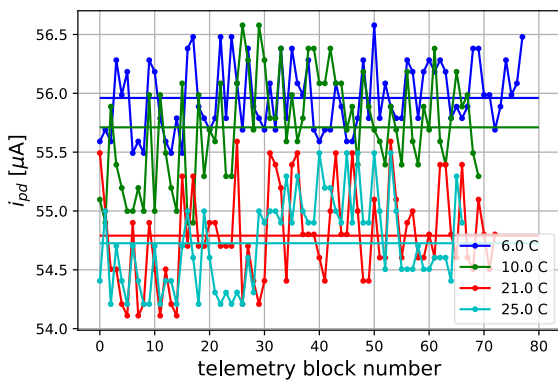


Fig. 4: Plot of the measured photocurrent for a fixed temperature value over time. Block number refers to the received telemetry block sequence number, which can be translated to mission elapsed time or standard Earth time.

Figure 5 shows a simplified equivalent circuit of the circuit used for the measurement of the photocurrent. The gain of the instrumentation amplifier has a value of $G = 991$. If V_{R2} corresponds to the voltage difference across $R2$, then $I_{pd} = V_{pd} / (991 * 49.9) \approx V_{pd} / 49451$. The voltage V_{pd} was digitized with an ADC featuring a resolution of 12 bits.

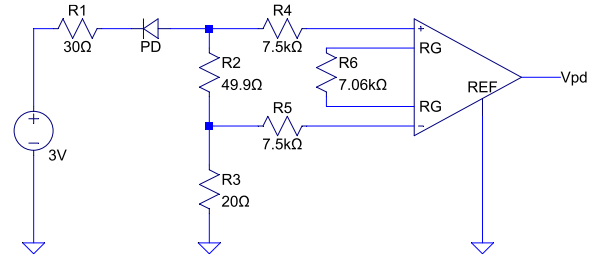


Fig. 5: Equivalent circuit used in the payload circuitry for the measurement of the photocurrent. PD is the integrated photodiode integrated in the payload PIC.

Assuming a photodiode responsivity value of $R = 0.88$ A/W, obtained from the photonic development kit (PDK) used to build the PIC, the optical power before the photodiode can be estimated as $P_{opt} = I_{po} / R$. Figure 6 shows the optical power received, averaged for each temperature value, and the least squares approximation line. Over the temperature interval [5, 30], the average optical power is -12.02 dBm, corresponding to a temperature of 16.5 °C. As expected, we can observe a reduction of the optical power as the temperature increases due to the reduction of the laser efficiency [12].

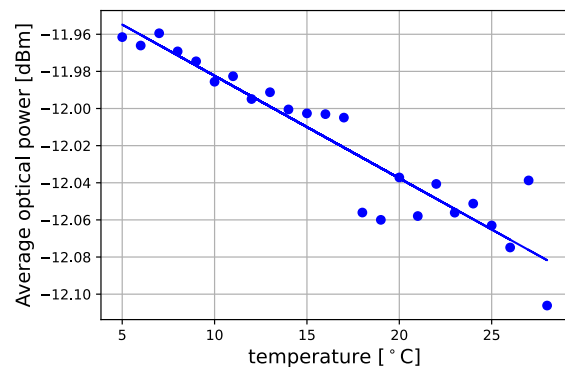


Fig. 6: Plot of the measured optical power, averaged for each temperature value.

Conclusions

In this work, we have demonstrated the field operation of an unshielded PIC consisting of a laser, an MMI coupler and a photodiode, onboard a picosatellite in a LEO sun-synchronous orbit. To the best of our knowledge, this is the first time that such a system has been demonstrated to date. The average measured optical power in the integrated photodiode was -12.02 dBm for a temperature of 16.5 °C.

The operation of the satellite over a year of collected data shows no degradation on the PIC operation due to ionizing radiation. This result looks promising for future missions, where similar PICs can be used for mmW/THz signal generation onboard low cost small satellites on LEO orbits with basic enclosures consisting only of the satellite framework PCBs.

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